

Solar + Battery Decision Simulator

Day-ahead and intraday market commitments under price and solar uncertainty.



Architecture plan | Estonia, post-Feb 2025 market regime

What the simulator does

A closed-loop environment: imperfect forecasts in, decisions out, realized profit at the end of each episode.

Price forecast

Day-ahead and intraday prices, with realistic error that shrinks as delivery approaches.

Solar forecast

Generation per period, with error correlated across nearby hours.

Asset state

Battery state of charge, throughput history, current realized solar.



Policy → decisions

Day-ahead commitments (quantities per delivery period)

Intraday trades (across discrete auction rounds)

Battery dispatch (charge / discharge per period)

Then time advances → realized solar materializes, prices clear, imbalance is settled, profit is recorded.

Scope and defaults

Two confirmed constraints from you, three defaults for the open choices. Local adjustments only if any default is wrong.

CONFIRMED BY YOU

Battery charges from solar only

No grid charging. Simplifies action space; removes a whole arbitrage branch.

All prices are simulated

Day-ahead, intraday, and imbalance prices come from a synthetic model — not historical replay.

DEFAULTS (CHANGE IF WRONG)

Configurable Market Time Unit

Quarter-hourly internal; hourly mode aggregates to 4-period blocks.

Discrete intraday auctions

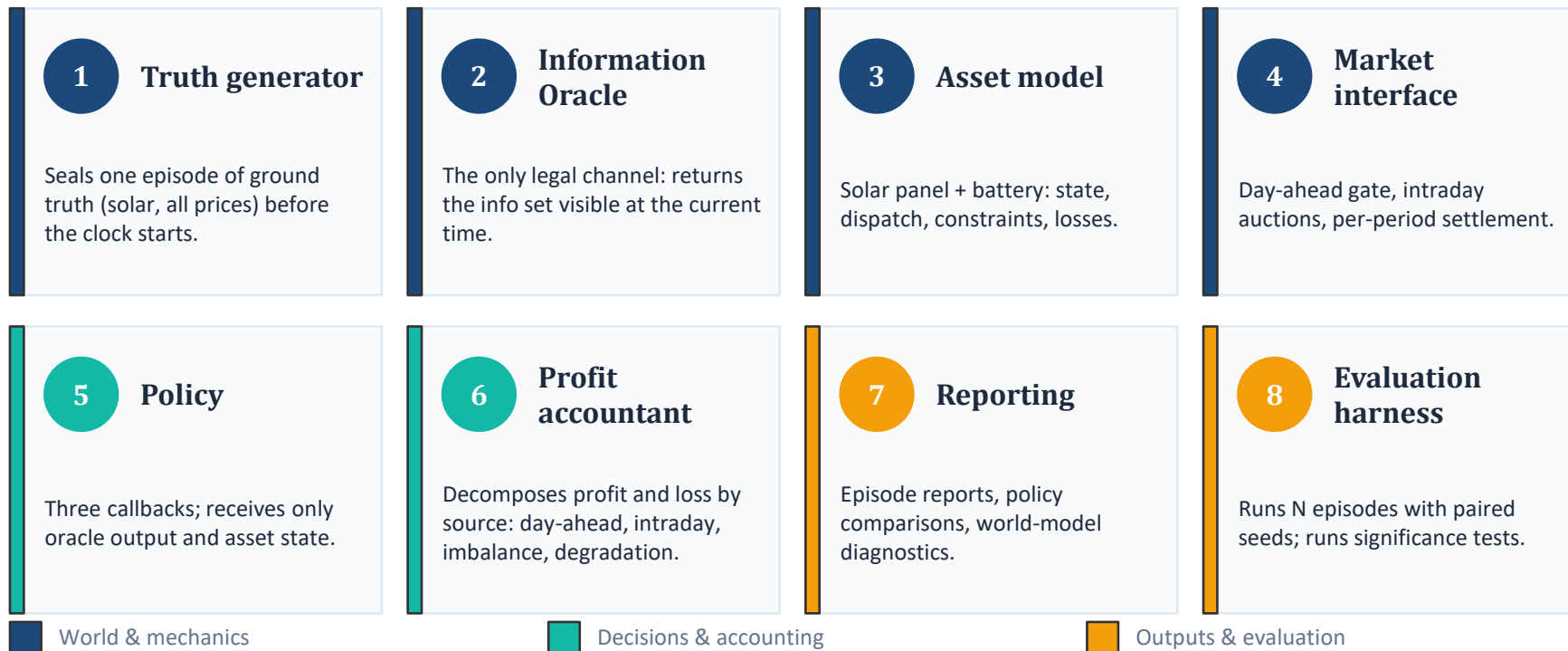
Fixed schedule per delivery period (e.g., 6h, 3h, 1h before delivery).
No continuous order book.

Flexible policy interface

Supports hand-designed, optimization-based (Pyomo/HiGHS), and learning-based policies.

Architecture: eight modules

Strict information firewall — the policy only ever sees what the Information Oracle returns at the current clock time.



Synthetic world: five-layer construction

Calibrated to Estonia post-Feb-2025 statistics, even though no individual day is replayed.

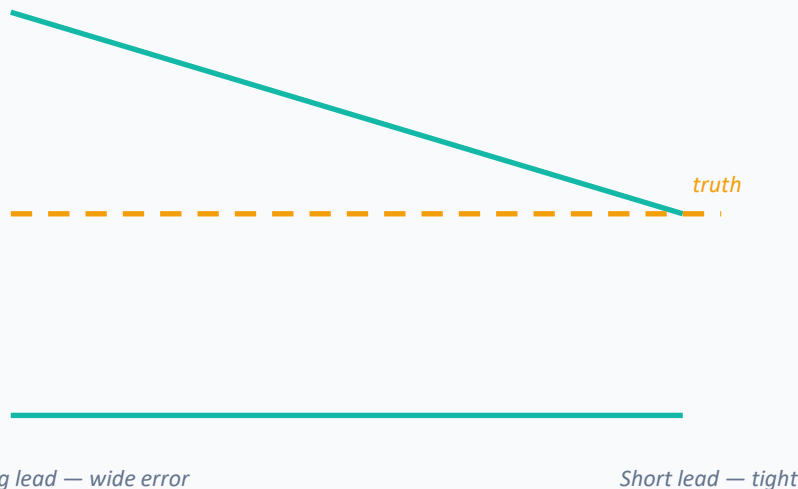
L1	Latent state	Weekly + annual seasonality and a slow regime indicator (calm / stressed) on a Markov chain.
L2	Solar generation	Latent clearness factor × diurnal envelope (day-of-year dependent), with cloudy-day shocks.
L3	Day-ahead price	Function of residual-load proxy, gas/CO2 state, regime, seasonality; plus a heavy-tailed shock.
L4	Intraday price	Day-ahead price + spread per auction round; spread innovations correlated with day-ahead shock direction.
L5	Imbalance price	Day-ahead + sign-dependent markup; correlated with aggregate solar shock so the operator's risk is real.

Critical: Layer 2 (solar) and Layer 3 (day-ahead) share the residual-load shock — so sunny days really do depress midday prices.

Forecast model — imperfect by construction

Forecasts emerge as a property of the world. Errors shrink with lead time and are correlated across nearby periods.

Forecast uncertainty vs. lead time



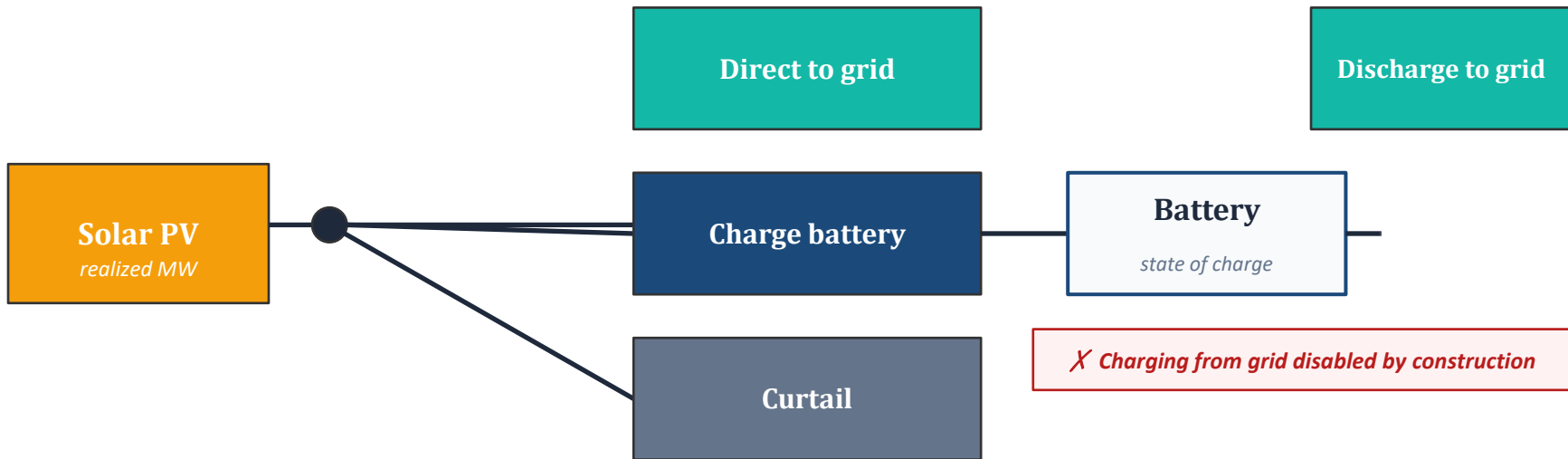
How forecasts are generated

- **Forecast = truth + structured error**
- Error variance scales with lead time
- Shocks correlated across nearby periods (not independent noise)
- Forecast updates approximate a martingale around truth
- Pluggable variants: perfect, good, biased, fat-tailed

Independent hourly noise would give the policy a free averaging gain that does not exist in reality. The shock structure prevents that.

Asset model — solar PV and battery

Battery is bounded by realized solar at each period. Forecast errors on solar translate directly into imbalance exposure.



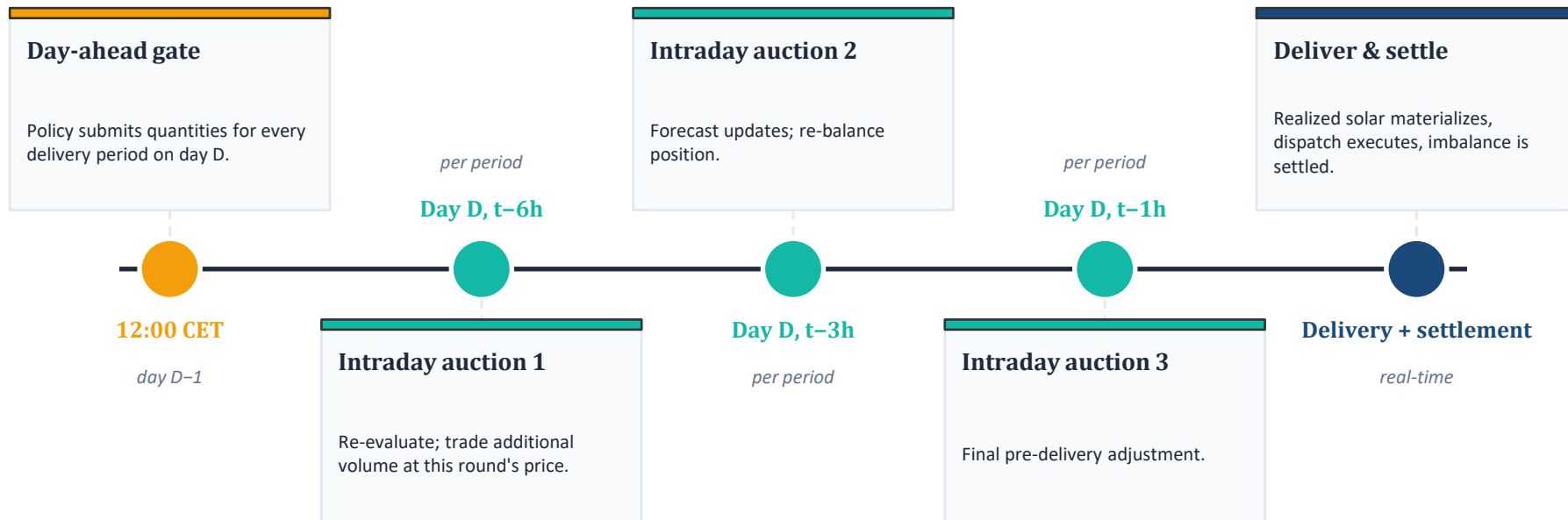
Key parameters

- Battery: capacity (MWh), max charge/discharge power (MW), round-trip efficiency, state-of-charge bounds, optional degradation cost.
- Solar PV: rated peak power, inverter clipping limit. Realized output comes from the world model.

Daily clock — gates, auctions, delivery, settlement

TIMELINE 08

Discrete events on a single delivery day. The policy is queried only when an event fires.



$$\text{Per-period profit} = \text{DA quantity} \times \text{DA price} + \sum (\text{intraday trades} \times \text{prices}) + (\text{actual} - \text{committed}) \times \text{imbalance price} - \text{degradation cost}$$

Policy interface — three callbacks, one class

Information comes only from the Oracle. The harness guarantees no look-ahead by construction.

```
class Policy:
```

```
    def decide_day_ahead(  
        info_at_noon, asset_state  
    ) -> day_ahead_quantities
```

```
    def decide_intraday(  
        info_now, asset_state,  
        position, auction_round  
    ) -> intraday_trades
```

```
    def decide_dispatch(  
        info_now, asset_state, position  
    ) -> battery_action
```

decide_day_ahead

Called once at 12:00 CET on day D-1. Sets quantities for all delivery periods on day D.

decide_intraday

Called at each scheduled auction round on day D. Trades on top of existing position.

decide_dispatch

Called per delivery period on day D. Sets the battery's charge/discharge action.

Optimization stack — Pyomo + HiGHS

One algebraic model serves both Model Predictive Control (rolling re-solve) and two-stage stochastic optimization.

Pyomo — modeling layer

- Algebraic model in Python
- Decision variables: dispatch, intraday trades, day-ahead quantities
- Constraints: state-of-charge dynamics, power limits, solar-only charging
- Switchable to mixed-integer if charge/discharge mode flags are needed

HiGHS — the solver

- Open-source linear and mixed-integer programming solver
- Fast, well-maintained, free to deploy at any scale
- Called from Pyomo via its appsi or solver-factory interface
- Same solver for every optimization-based policy

Model Predictive Control

Re-solve the deterministic problem at each decision point using the latest forecast. Implement first decision, roll forward.

Two-stage stochastic optimization

At the day-ahead gate, optimize across scenarios drawn from the policy's own forecast model. Recourse: intraday + dispatch.

Reporting module — what comes out

Four report families. Every result is reproducible from configuration + random seeds.

Per-episode reports

- Profit decomposition: day-ahead / intraday / imbalance / degradation
- Time series: prices, solar, state of charge, committed vs. realized
- Per-decision diagnostics (forecast at that moment, action taken)
- Output: HTML or PDF, one per episode

Policy comparison

- Profit distributions across episodes (paired seeds, same world)
- Regret vs. perfect-foresight upper bound
- Tail metrics: Conditional Value at Risk, worst-day loss
- Paired statistical tests on daily profit differentials

World-model validation

- Marginal distribution checks (prices, solar, imbalance)
- Cross-correlations: solar–price, intraday–day-ahead, imbalance direction
- Tail and spike frequency vs. historical reference
- Pass/fail gate for Phase 3 (calibration)

Run summary & reproducibility

- Configuration snapshot (world model, forecast model, policy)
- Random seeds, solver versions (Pyomo, HiGHS)
- Runtime, episodes completed, errors
- Output: machine-readable manifest plus human summary

Phased build

Each phase has a single acceptance test. Move forward only when the gate passes.

1	Deterministic skeleton	Asset, market, settlement, profit accounting. Constant prices and solar.	<i>Profit matches hand calculation for a known day, to the cent.</i>
2	Synthetic world v0	Joint world model with simple correlations. Gaussian forecast errors. Benchmarks.	<i>Perfect-foresight > Model Predictive Control > naive, paired across seeds.</i>
3	Synthetic world v1 (calibrated)	Calibrate to Estonia post-Feb-2025 data. Realistic cross-correlations.	<i>All five world-validation statistics fall within plausible bands.</i>
4	Optimization policies (Pyomo + HiGHS)	Model Predictive Control and two-stage stochastic optimization behind the same Pyomo model.	<i>Stochastic optimization beats Model Predictive Control on average and on tail risk.</i>
5	Extensions	Sensitivity studies, 15-minute mode, reinforcement-learning interface, richer imbalance pricing.	<i>Any extension promoted only if it beats Phase 4 baseline on the same metric.</i>

Phase

Goal

Acceptance gate

Validation and main risks

Two failure modes are far more damaging than picking the wrong policy class.

Phase 3 validation suite

- Day-shape autocorrelation of prices
- Distribution of intraday–day-ahead spreads
- Imbalance price vs. system direction
- Midday solar–price correlation
- Tail behavior — negative prices and spikes

Right marginals, wrong correlations

World model passes marginal checks but breaks the solar–price link. Policy comparisons end up dominated by an unrealistic hedge.

Forecast errors too kind

Independent or weakly correlated forecast errors give the policy free averaging across periods. Hedging looks better in simulation than it ever is in reality.

Mitigation: over-invest in the Phase 3 validation suite. Every later result depends on it.

Open questions

OPEN 14

Three answers would sharpen Phase 4 substantially. They are not blockers for Phases 1–3.

Q1

Battery energy-to-power ratio?

A 1-hour battery and a 4-hour battery support very different policies — short-duration is mostly imbalance hedging, long-duration enables real arbitrage across the daily price shape.

Q2

Target episode count per experiment?

Determines whether the world model must be vectorized from day one (thousands of episodes) or can stay iterative for clarity (hundreds).

Q3

Calibration period for the synthetic price model?

Estonia post-9-Feb-2025 only (the in-regime window) gives realism but few samples; including 2024 adds samples but mixes regimes. Pick before Phase 3.